



# Apoorv Pandey

B.Tech Computer Science — AI & Data Science — 2027

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## Education

B.Tech in Computer Science (AI & Data Science)

BML Munjal University

2023 – 2027

CGPA: 6.2 / 10

## Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Bash

**Backend & Web:** FastAPI, Flask, Next.js, React, Pydantic, SQLAlchemy, Celery, Prisma, Tailwind CSS

**AI / ML:** LLM Integration, Prompt Engineering, RAG, Agentic AI, LangChain, LangGraph, CrewAI, vLLM, Azure OpenAI, Sentence-Transformers, Pandas, NumPy, scikit-learn

**Infrastructure & DevOps:** Docker, Docker Compose, Nginx, Cloudflare Tunnel, Linux, Azure App Service, Azure Container Apps, GitHub Actions, CI/CD

**Data & Systems:** PostgreSQL, Redis, S3, WebSocket/Socket.IO, Server-Sent Events, REST APIs, JWT, Microservices, Rate Limiting, Async Programming

## Selected Projects

**AI/LLM Clinical Text Analysis Platform (SNAP-AI)**

Mar 2026 – Present [GitHub]

*Python, FastAPI, Celery, vLLM, GPT-OSS-120B, PostgreSQL, Redis, NVIDIA DGX A100*

- Architected a 3-layer clinical NLP pipeline extracting postoperative complications from German surgical discharge summaries, assigning Clavien–Dindo grades, and computing the Comprehensive Complication Index (CCI®).
- Engineered a fault-tolerant Celery batch system with idempotent skip-completed logic and resume support; processed **156 real clinical cases** unattended on a cluster of **4 NVIDIA A100-40GB GPUs** and achieved **107/107 backend test passes**.

**AI Business Knowledge Assistant**

2026 – Present [GitHub]

*Python, FastAPI, Next.js, RAG, BM25, RRF, Cross-Encoder, Claude, SQLite, Docker*

- Built a multi-source retrieval and orchestration engine with **4 query paths** (PDF, SQL, hybrid, insufficient-evidence), producing grounded answers with citation verification and evidence provenance.
- Implemented hybrid retrieval using dense embeddings + BM25, **Reciprocal Rank Fusion (RRF)**, cross-encoder reranking, schema-aware SQL generation, sqlglot validation, and read-only query constraints.
- Designed an inspector-first UI exposing routing decisions, retrieved evidence, generated SQL, retrieval scores, citation verification, latency, and cost; supports **English/Hebrew** and deterministic offline execution.

**AI Account Coding Engine**

Dec 2025 – Jan 2026 [GitHub]

*Python, FastAPI, Azure OpenAI (GPT-4), Sentence-Transformers, Docker, Azure*

- Built and delivered a production AI microservice automating General Ledger account coding for German-language construction/manufacturing invoices across **3 milestones**.
- Architected a hybrid classifier cascading regex rules, GPT-4 few-shot inference, and multilingual semantic embeddings; achieved **50–100 ms rule-path latency**, supported **100 line items/request**, and degraded gracefully to rule-only inference when Azure OpenAI was unavailable.

**Client Hunter Agent**

2025 [GitHub]

*Python, Multi-Agent Systems, BeautifulSoup, Playwright, Gemini Vision, OpenAI, Apollo, Google Sheets/Airtable*

- Developed a **5-agent** B2B lead-research workflow covering orchestration, adaptive scraping, decision-maker enrichment, personalization, and structured output to Google Sheets/Airtable.
- Added 3-tier scraping fallbacks, circuit breakers, rate limiting, and graceful degradation; tested across **500+ candidate profiles** with **0 decision-maker false positives** in a 50-law-firm evaluation.

## Internship

**Avrio Energy Pvt. Ltd., Gurugram — AI Prompt Engineering & Data Automation** Jun 2025 – Aug 2025

*Data Science & Python Intern*

- Integrated LangSmith API to design, optimize, and trace high-performance AI prompts for bill and document extraction.
- Automated workflows with Google APIs for location-based analytics in building creation and asset onboarding; built, trained, and deployed ML models and custom annotation APIs for equipment detection.
- Developed multilingual PDF parsing for invoice/bill extraction and designed RESTful APIs with Swagger UI and Pytest; delivered scalable batch frameworks with error handling, confidence scoring, and structured logging.

## Certifications

**IBM Generative AI Engineering Professional Certificate** — Coursera, Dec 2025

[Credential]